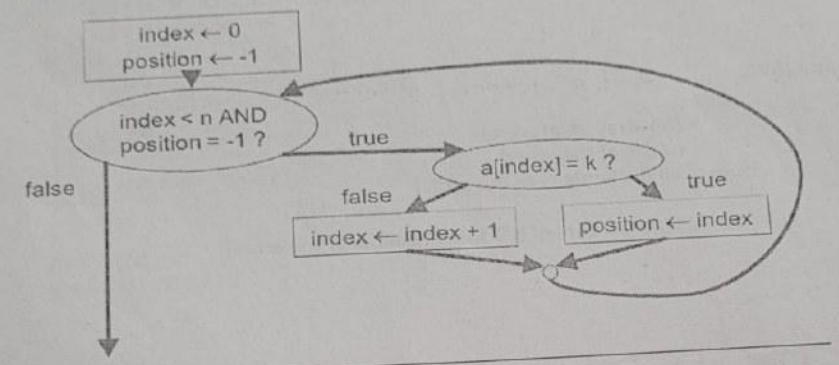
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TOOLS OF PROGRAMMING EXAM
PROF. BONIFACE MBOUZAO
TIME: 3H00

QUESTION 1

Translate this algorithm that find an integer k in an array into JAVA. Here, if K is in array a, the algorithm return its position and if K is not in the array the algorithm retruns -1.

GIVENS : a (an array of integers)
n (SIZE OF THE ARRAY)
k (value to find the position in the array)
INTERMEDIATES :
index (index to go through the array from 0 to N-1)
RESULT : position (position of k in the array)
HEADER : position <---------- whereisk(a,n,k)
BODY :



QUESTION 2:

This is a array of length 4, answers to the reference to this array a:

2	-1	5	7	(length 4)

a[2] =
a[k] =
a[2*k-1] =
a[a[0]+1] =

QUESTION 3:

Trace this algorithm and tell what this algorithm is doing

```
class SwapTilYouDrop
{
  public static void main (string args[ ])
  { int i = 0;
    int a;
    a = new int []
                  { 2, 4, 6, 8, 10, 12 } ;
    while( i <= 2 )
    {
      arraySwap(a, i, 5 - i ) ;
      i = i + 1;
    }
    for ( i = 0 ; i <= 5 ; i = i + 1 )
    { System.out.println("a[" + i + "] is "
      + a[i] );
    }
  }
  // arraySwap : swaps values at i,j
  // Givens: x, an array,
  //         i,j, 2 indices in x
  public static void arraySwap(int[ ] x,
                               int i,int j)
  {
     // DECLARE VARIABLES
     int temp ; // Inter: holds x[i]
     // BODY OF ALGORITHM
     temp = x[i] ;
     x[i] = x[j] ;
     x[j] = temp;
  }
```

QUESTION 4

Translate this algorithm into JAVA.

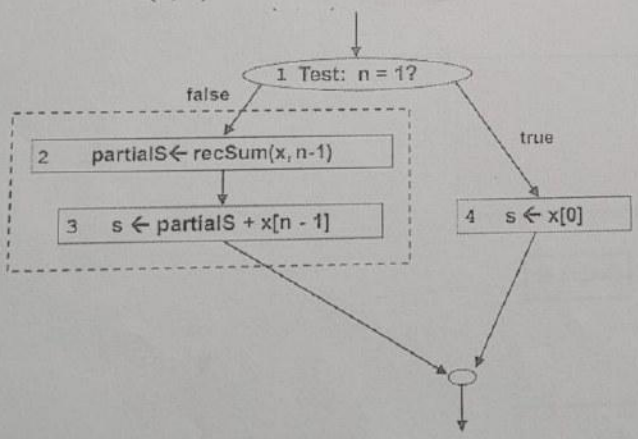
GIVENS : x (An array of interger)
n (size of the array)

RESULT : s (sum of all the lements of the array)

INTERMEDIATE :
partialS (Partial sum of N-1 first elements of the array)

BODY :

s ← recSum(x, n)



Question 5

Give the algorithm of this Method and give the Givens (Data), Result, Intermediate, Header and Body of the algorithm.

```
public static void locateLargestValue(int[] a, int n)
{
  // VARIABLE DECLARATIONS
  // GIVENS: a - reference to array to sort
  //         n - number of elements to sort - note
  //             that a.length CANNOT be used.
  int pos;  // RESULT - position of largest value
  // INTERMEDIATES
  int m;    // smaller
  int posM; // used for swapping

  if(n == 1)
  {
     pos = 0;  // base case
  }
  else
  {
     m = n-1;
     posM = locateLargestValue(a, m);  // recursion
     if(a[posM] > a[n-1])
     {
       pos = posM;
     }
     else
     {
       pos = n-1;
     }
  }
  // RESULT
  return(pos);
}
```